

$$\tau_p L\left(\frac{dy}{dt}\right) + L(y) = K_p L(f(t))$$

$$\Rightarrow \tau_p [sL(y) - y(0)] + L(y) = K_p L(f(t))$$

$$\Rightarrow L(y) = \frac{K_p L(f(t))}{s\tau_p + 1} \rightarrow \textcircled{1}$$

$$\Rightarrow \text{Transfer function} = \frac{K_p}{s\tau_p + 1} \rightarrow \textcircled{2}$$

↳ Ratio of Output / Input of

Laplace Transform.